

Transformer MLP Gate Thresholds Are Couplings to a Carried Reference Direction

Olli Tuomi
Evident Solutions Oy

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Abstract

The corpus-mean direction of a transformer’s residual stream is a component shared across all inputs, and is commonly removed by mean-centering before representational analysis. We present evidence that it is a functional component: the reference against which the MLP gate population sets its operating point. In Phi-2, an exact decomposition of resting gate pre-activations shows that at mid-stack layers the resting inhibition of $>99.9\%$ of gates is carried by the coupling $w \cdot b_L$ to the carried mean direction, at $48\text{--}56\times$ the explicit bias parameter, and the coupling is direction-specific: a random direction at matched norm orders the population’s firing rates at Spearman $\rho \leq 0.09$ (median over 50 draws; up to 0.22 in the tail) where the reference reaches 0.95. (That 0.95 is near-tautological on its own; §5 derives its null.) Causally, removing the stream’s projection on the reference multiplies above-threshold firing by $\sim 9\times$, dose-monotonically, at $23\text{--}59\times$ a norm-matched control; a random-initialised twin is flat, and replacement tests show that the direction carries the function and the magnitude does not. A direction-matched control makes the same point: noise injected along the reference costs $8\text{--}42\times$ the same energy along a random direction. The decomposition replicates on four further families spanning both gate types (GELU with an explicit gate bias, bias-free SwiGLU), and the dose-response on two of them. Across an eight-model scan the mechanism is present in every GELU and SiLU family and absent only in OPT, where an opposing LayerNorm bias cancels the carried reference. Gate thresholds are implemented as couplings to a constant the network builds for the distribution it is reading, with the bias parameters contributing little; how much of that constant is carried in the stream and how much in parameters depends on the architecture.

1 Introduction

Transformer residual streams contain large, stable components that are shared across inputs: massive activations (Sun et al., 2024), which give rise to attention sinks (Xiao et al., 2023; Sun et al., 2024), and outlier dimensions (Kovaleva et al., 2021; Timkey and van Schijndel, 2021), whose magnitude tracks token frequency (Puccetti et al., 2022). Existing work locates these objects and shows they matter for quantisation and attention. What they are *for*, computationally, is less settled.

This paper studies one such object, the corpus-mean direction of the layer-normalised residual stream, b_L , and measures a specific function for it: it is the reference against which the MLP gate population sets its operating point. The claim decomposes into parts that were tested separately, each against a random-initialised control model (“twin”) of identical architecture:

1. **Object** (§2): b_L is constant across text for a given input distribution, is held at a fixed fraction of the growing stream, and, in Phi-2, rotates slowly with depth as a band of related directions.

2. **Function** (§3): removing or amplifying the stream’s projection on b_L releases or suppresses gate firing, dose-monotonically, far beyond norm-matched controls; the twin is flat.
3. **Identity** (§4): the function is carried by the direction, not by magnitude.
4. **Implementation** (§5): resting gate inhibition is the coupling $w \cdot b_L$, not the bias parameter. This is an exact decomposition, not a fit.
5. **Persistence and dependence** (§6): the direction is continuously re-supplied by aligned per-layer mean writes, so it re-accumulates after local removal; defeating that restoration by co-removal is near-catastrophic for prediction.
6. **Development** (§7): the operating point is built early: on Pythia-410M it forms in the induction-transition window, a finer 160M grid shows it forms first, and it forms even with attention disabled (one 160M seed), so the token-local prediction task alone is enough to build it.

These experiments are run on Phi-2 (Abdin et al., 2023), with Development on Pythia-410M (Biderman et al., 2023) checkpoints and two small trained models. The two most portable tests, the static decomposition (§5) and the causal dose-response (§3), are then tested on further families; the remaining causal tests are Phi-2-only (Limitations).

Throughout, “gate” means the scalar nonlinearity of an MLP hidden unit; “duty cycle” is the fraction of positions at which a unit’s activation is positive. Two facts motivate the hypothesis. First, Phi-2’s median duty cycle over 400 sampled gates is 0.076, against 0.48 at random init: trained gates are off by default, in line with the known activation sparsity of trained transformer MLPs (Li et al., 2022; Szatkowski et al., 2025), and “off by default” is only defined relative to what occupies the stream at rest. Second, take a gate’s pre-activation at the positions where it fires most strongly and ask what share of it comes from the stream’s corpus-mean component (the *reference share of drive*; negative values mean that component opposes firing). Its median over gates is about -0.40 pooled over the eight layers measured (-0.31 to -0.55 per layer), and ≈ 0 in the twin (-0.11 to $+0.01$): trained gates are held down by something constant. The experiments below ask whether that constant is functional structure.

Models. Primary model: Phi-2 (Abdin et al., 2023) (2.7B, 32 layers, width $d=2560$), fp16, hooks on public weights; twin: same architecture, random init. Cross-model replication: GPT-2-medium (Radford et al., 2019), Pythia-1.4B (Biderman et al., 2023), Qwen2.5-1.5B (Qwen et al., 2024), and TinyLlama-1.1B (Zhang et al., 2024) carry the static decomposition, and Pythia-1.4B and Qwen2.5-1.5B also the causal dose-response; the static decomposition further runs across an eight-model GELU/ReLU/SiLU scan comprising those four plus Pythia-410M, SmoILM2 (Ben Allal et al., 2025), and two OPT models (opt-350m, opt-1.3b) (Zhang et al., 2022). Each is compared against its own twin (same configuration, random weights). Development (§7): Pythia-410M-deduped public checkpoints (Biderman et al., 2023), plus a 160M model (Pythia-160M architecture, standard GELU MLP) trained here on the Pile with dense early checkpoints, and an attention-disabled variant of it (each layer’s attention output projection zeroed and frozen).

2 The object

Definitions. m_L is the corpus-mean raw residual at the input of layer L ; b_L is the corpus mean of x_{in} , the layer-normalised MLP input at L (the object gates actually read); $\hat{m}_L$ the unit vector of m_L . The

two are closely aligned ($\cos(b_L, m_L) = 0.81\text{--}0.83$ at L6–L18), so manipulations along $\hat{m}_L$ act on the reference. Statistics are computed over document positions ≥ 16 , excluding the attention-sink region.

Constancy. Split-half cosine of b_L across document halves is 0.94–0.99 at every measured layer. The constancy holds for a given input distribution: the cosine between the reference estimated on prose and on other inputs is 0.97–0.98 for held-out prose, 0.76–0.84 for code, and 0.02–0.05 for a repeated token, where the twin keeps one direction for prose, code and random tokens (≥ 0.99) and only partly separates the repeated token (0.51–0.52). The mean also hides some finer structure: token classes such as digits, sub-word continuations and word-initial tokens have their own means, separable from b_L above a split-half floor and partly as a trained property, but the class partition accounts for only 2–4% of the variance around the mean.

A constant fraction of a growing stream. The residual stream grows several-fold across the stack, so a fixed coupling could only give a fixed operating point if something normalised it. Over L2–L28, mean $\|x\|$ at the layer input grows $25.9 \rightarrow 131.9$ ($5.1\times$) and $\|m_L\|$ grows $8.7 \rightarrow 51.1$ ($5.9\times$), so the mean’s *share* of the stream stays in 0.33–0.44 throughout; the pre-norm mean grows with the stream. LayerNorm then divides that growth out, and the object gates read is flat, $\|b_L\| = 6.6\text{--}10.1$ over the same span. The resting term (each gate’s pre-activation at the mean input, $w \cdot b_L + b_n$; §5) inherits the stability directly (median -1.57 at L6 to -1.19 at L28, within a band of about 0.7 while the stream quintuples). The twin separates this as a trained property: there the share is not stabilised at all but climbs monotonically ($0.54 \rightarrow 0.78$ over the same layers) and $\|b_L\|$ climbs with it ($27.2 \rightarrow 39.6$). What the network maintains is therefore a constant *fraction* of the stream: the pre-norm magnitude grows, and the post-norm operating point is the invariant.

What holds the fraction. The measurements above show this stability; an exact decomposition explains it. Writing $x_{\text{in}} = b_L + \delta$, the reference component is $\langle x_{\text{in}}, \hat{b}_L \rangle = \|b_L\| + \langle \delta, \hat{b}_L \rangle$, and because b_L is the corpus mean the second term has mean zero by construction: only its variance is at issue. Two things suppress it. First, LayerNorm’s per-position denominator: recomputing every contribution with σ frozen at its corpus mean raises the variance of the reference component from 0.62/0.51/0.96 to 4.25/2.28/2.83 at L8/L16/L24, so the per-position gain control removes 85%/77%/66% of it. Second, aggregation: the remainder is a sum over the 17–49 writes upstream of the layer, each contributing a standard deviation of at most 0.33 against a total mean near 6.7, and the sum is far steadier than its parts (coefficient of variation 0.118 against 0.25–6.65 for individual sources). The sources do not cancel each other: their contributions are weakly positively correlated (mean pairwise correlation +0.05), and covariance adds 34–72% of the total variance. The stability therefore comes from normalisation and averaging. The same decomposition has two by-products. The largest and steadiest single contributor is the layer-0 block: its attention and MLP writes supply 28% of the reference component at L8, with mean-to-spread ratios of 3.1 and 4.0 against 1.3–1.5 for late writes. And aggregation de-correlates as well as stabilises: a held-out ridge regression on twelve surface text features predicts the largest individual contribution at $R^2 = 0.34$ but the total at only 0.11 (permutation null -0.001), so the total tracks surface text far less than its parts do.

A slowly rotating band of directions. $\cos(\hat{m}_{10}, \hat{m}_{18}) = 0.94$, and the reference gates actually read is aligned over the same span ($\cos(\hat{b}_{10}, \hat{b}_{18}) = 0.92$). Extending the comparison to every layer shows a band: layers four apart sit at ≈ 0.97 (adjacent ones at 0.99), but $\cos(\hat{b}_6, \hat{b}_{22}) = 0.81$ and $\cos(\hat{b}_6, \hat{b}_{30}) = 0.24$. The coupling survives the drift: substituting another layer’s reference into layer L ’s resting term, at layer

L 's own magnitude, still orders layer L 's duty at Spearman $\rho = 0.86\text{--}0.96$ anywhere mid-stack and at $0.69\text{--}0.77$ even from L30, against $0.06\text{--}0.09$ for a random direction at matched norm (§5; this contrast carries the result, since ρ alone is near-tautological); in the twin the same transfer instead tracks the cosine almost exactly ($0.44 \rightarrow 0.41$, $0.71 \rightarrow 0.70$), so the tolerance is trained. Causally the same point is made in §4: layer 10's gates accept layer 18's mean direction as their reference almost without loss. This cross-layer geometry is specific to Phi-2 and varies across the replication families (Limitations); every other claim in the paper is per-layer and stands without it.

Content. Decoded through the *final* unembedding, mid-stack b_L (L2–L18) aligns only weakly with any single token direction (max row cosine $0.14\text{--}0.15$, about $1.5\text{--}1.7\times$ the isotropic floor), rising toward the readout. This paragraph is descriptive; no claim in §3–§7 depends on it.

Sources. The embedding mean is negligible (norm 0.42 , cosine 0.02 to the deep mean, m_L at L30). Instead, every layer's attention *and* MLP mean writes are mutually aligned with the deep mean (cosines $+0.10$ to $+0.77$ through L27, never negative), each small (norms $\sim 1\text{--}4$ through L27, against a mean norm of $9\text{--}51$; the last four layers write larger and less consistently aligned), summing coherently: the reference is a distributed construction to which every component contributes a small aligned constant term. Part of that constant is input-independent: the module biases and the LayerNorm bias supply a share that depends on the architecture ($41\text{--}47\%$ in Phi-2; §5).

3 Function: the gate population thresholds against it

Manipulation. At layer $L \in \{6, 10, 14\}$, the residual entering the layer is modified at all positions ≥ 16 : $x' = x + (\alpha - 1)(x \cdot \hat{m}_L) \hat{m}_L$, for $\alpha \in \{0, 0.5, 1, 1.5, 2\}$. $\alpha=1$ is the identity (verified as an exact no-op: the output distribution is unchanged); $\alpha=0$ removes the stream's projection on the reference; $\alpha=2$ doubles it. The control subtracts the same per-position magnitude along a fixed random unit direction (norm-matched). Firing is measured as $\text{frac}(\text{act}>0.5)$, the fraction of post-GELU gate activations above 0.5 , and output damage as the KL divergence of the final-position next-token distribution from the clean run's. 39 documents.

Result. Table 1 shows the dose-response. Removal releases firing; amplification suppresses it; the response is monotone through the identity point at every layer and $23\text{--}59\times$ the norm-matched control, consistent with near-binary gates driven across threshold together, as reported for GPT-2's MLP units (Balogh, 2026). Against the clean baseline of $\text{frac}(\text{act}>0.5) = 0.0105$ at L10, the release is a $\sim 9\times$ multiplication of above-threshold firing. The suppression side saturates small, which is the expected asymmetry: at duty 0.076 the population is already near its inhibited floor, so added inhibition has little left to turn off while release frees every marginal gate. The twin shows no response, having no resting coupling to lose (§5).

Output damage. Removal at a single site costs $\text{KL} = 1.43/0.52/0.27$ (L6/L10/L14) against $0.020\text{--}0.025$ for the norm-matched control. The twin loses only $0.32\text{--}0.38$ KL under the same removal despite its raw mean being $8\text{--}10\times$ larger in norm ($121\text{--}208$ vs. $15\text{--}21$): per unit of removed norm the trained model is roughly two orders of magnitude more sensitive.

α	$\Delta\text{frac}(\text{act}>0.5)$			KL vs. clean		
	L6	L10	L14	L6	L10	L14
0.0 (removed)	+0.103	+0.086	+0.075	1.427	0.519	0.274
0.5	+0.023	+0.023	+0.021	0.047	0.040	0.035
1.0 (identity)	0.000	0.000	0.000	0.000	0.000	0.000
1.5	−0.006	−0.007	−0.007	0.014	0.018	0.020
2.0 (doubled)	−0.008	−0.009	−0.009	0.055	0.060	0.110
norm-matched control	+0.0028	+0.0015	+0.0033	0.020	0.022	0.025

Table 1: Dose-response of gate firing and output damage to scaling the stream’s projection on $\hat{m}_L$ at the manipulated layer (Phi-2, trained). The twin is flat at every α ($|\Delta\text{frac}| \leq 0.003$, no ordering). Point estimates are document medians over 39 documents; 95% document-bootstrap CIs (2000 re-samples) on the removal row: Δfrac $[+0.096, +0.112]/[+0.079, +0.098]/[+0.068, +0.088]$ and KL $[1.14, 2.03]/[0.36, 0.75]/[0.15, 0.40]$ (L6/L10/L14), each disjoint from the norm-matched control’s KL CI ($\leq [0.010, 0.032]$). The document-sampling spread on KL dwarfs the run-to-run fp16 jitter ($\sim 1\%$), so only the leading two digits are informative.

A matched-energy scale. Two matched-energy controls place the removal cost. Isotropic noise injected at one layer carrying the reference’s own share of the stream (norm share 0.33, §2; about 10% of the energy) costs KL 0.013 at both L6 and L10, against 1.68 (L6) and 0.69 (L10) for removing b_L on the same per-position measure (a mean over positions, distinct from Table 1’s final-position KL): the reference removal is $127\times$ (L6) to $52\times$ (L10) more damaging than matched-energy noise, and $670\text{--}4200\times$ more than removing a random direction. Turned around, injecting per-position noise *along* $\hat{b}_L$ costs $8\text{--}42\times$ the same energy injected along a matched random direction, across layers and amplitudes; enforcing a near-zero empirical mean on the injection (residual $\|\langle n \rangle\|/\|b_L\| \leq 0.01$ in three of four settings, 0.06 in the fourth) leaves the ratio unchanged, so the cost comes from per-position jitter on the reference axis and not from a shift of the operating point. Both corroborate the direction specificity of Table 1 by an independent route, measured on KL.

Where the damage lands. Resolving the same removal by context locates the cost. Applying it at L16 on a 16-sentence corpus costs $\Delta\text{NLL} = +0.493$ (change in next-token negative log-likelihood) and $\Delta H = +0.379$ (change in output entropy) while keeping top-1 at 0.70, and the loss is concentrated: $+0.94$ at high-entropy positions against $+0.04$ at low-entropy ones, and $+0.88$ at function-word positions against $+0.36$ at content words. Replacing the massive-activation coordinates with their corpus constants at the same site is a much milder intervention on the same measure ($\Delta\text{NLL} = +0.086$, top-1 0.87), and its damage sits in the same places. Removing the reference therefore costs most at the high-entropy, least-certain predictions and little where the next token is already near-settled.

An input-side demonstration, and what supplies the reference. Every manipulation above is an activation edit, which leaves open the objection that edited states are off the distribution the model ever visits (Mishra et al., 2026). The same conclusion follows from a real input with no edit: feeding a degenerate input (one token repeated) inflates LayerNorm’s per-position denominator by $8\text{--}15\times$, which divides the reference out: the reference component falls from 6.8 to 0.9 at L8 and from 7.1 to 1.0 at L24, going slightly negative in the most degenerate condition. Because the coupling is inhibitory, removing it releases firing, and duty rises from 0.087 to 0.791 at L24 (a $9\times$ increase) with the weight-space resting term’s correlation with measured mean pre-activation falling from 0.95 to 0.16. This is the same release

the α scaling produces, obtained from a real prompt. Repeated-token input is also known to disrupt the attention-sink circuit, with a small set of early-layer MLP neurons raising the hidden-state norm of the repeated tokens (Yona et al., 2025); that is consistent with the inflated LayerNorm denominator measured here, and the measurement here concerns its effect on the MLP gates.

Two scope statements belong with it, because they bound how much the model depends on its input distribution. The failure needs extreme degeneracy: tiling the first k tokens of a document, the denominator is inflated at $k = 1$ and $k = 2$ but is already back to its prose value at $k = 4$ ($\sigma = 15.2/2.7/0.92$ against a prose 0.99), and an i.i.d. draw from four distinct token types is enough. And recovery is local: appending a clean prompt after a degenerate context brings duty most of the way back within two or three tokens ($0.739 \rightarrow 0.542 \rightarrow 0.207 \rightarrow 0.186$ by offset at L24, against a prose 0.08). Natural text is therefore nowhere near the failure regime, and a short window of ordinary recent context is enough to restore the reference.

Reach in sequence position. A single-site removal is local in depth (§6) but reaches far along the sequence: removing the reference in an eight-position window at L6 and then reading only *untouched* later positions, the next-token distribution is still altered 32 tokens past the window at $KL = 0.098$, 16% of the in-window magnitude and $54\times$ the norm-matched control, and the ratio to control *grows* with distance, since the control decays faster than the effect. Greedy continuations diverge from the unperturbed ones within a median of two steps. The effect alters later predictions without making them worse on average: KL is large, but the change in the true token’s log-probability is inconsistent in sign.

Cross-model. The same manipulation at one mid-stack layer (depth ≈ 0.31 : Pythia-1.4B L8, Qwen2.5-1.5B L9; 29 documents, each against its own twin) reproduces the sign, the monotonicity, and the control separation. On the duty-cycle measure (activation > 0), removal shifts duty by $+0.318$ [0.303, 0.329] (Pythia) and $+0.302$ [0.286, 0.316] (Qwen), doubling by -0.124 and -0.081 , monotone across a 15-point α grid, against $+0.0001$ for the norm-matched control and flat twins ($|\Delta\text{duty}| \leq 0.002$); Phi-2 on the same measure is $+0.238/-0.070$. Qwen is the sharpest case: its SwiGLU gate has no bias parameter, so the carried reference is the entire resting inhibition, and scaling it still moves firing in dose-response. Removal costs final-position KL 3.60 (Pythia) and 0.59 (Qwen). Weakening the reference also raises output entropy in all three models tested (ΔH at $\alpha=0.8$ is $+0.057$ Pythia, $+0.047$ Qwen, $+0.013$ Phi-2, each above a norm-matched control with $|\Delta H| \leq 0.0005$), with the minimum at the natural magnitude in Phi-2 and Pythia (in Qwen the amplification side is noisy and dips below it).

4 Identity: the direction, not the magnitude

Removal tests necessity of the projection, but it leaves two alternative readings open. Gates might not read this direction at all and simply need the stream to be full; and since the manipulation acts on the pre-norm residual, removing the projection shrinks the norm entering LayerNorm, and the renormalisation then boosts the gain on every orthogonal component; some of the release could come from that gain boost. The replacement panel addresses both: at L10, $x' = x - (x \cdot \hat{m}_{10})\hat{m}_{10} + (x \cdot \hat{m}_{10})\hat{c}$ transfers the signed per-position projection onto a candidate direction $\hat{c}$; magnitude, sign pattern, and timing are preserved, the total norm is approximately unchanged (the cross-terms with $\hat{c}$ are small), and only the direction changes.

In Table 2, shuffled and random directions release firing almost exactly like removal; a direction at cosine 0.096 to $\hat{m}_{10}$ buys a sliver proportionate to its cosine and nothing more; and another layer’s mean direction (cosine 0.94) preserves baseline firing almost perfectly, $31\times$ less release and $130\times$ less KL than removal. The magnitude-reader alternative is excluded: no low-cosine candidate preserves anything. The

replacement $\hat{c}$	$\cos(\hat{c}, \hat{m}_{10})$	$\Delta\text{frac}(\text{act}>0.5)$	KL
none (pure removal)	—	+0.0845	0.516
coordinate-shuffled $\hat{m}_{10}$	−0.003	+0.0840	0.950
random direction	+0.011	+0.0778	0.484
function-word W_U sum	+0.096	+0.0740	0.395
layer 18’s $\hat{m}_{18}$	+0.940	+0.0027	0.004
identity	1.000	0.000	0.000

Table 2: Replacement panel at L10 (Phi-2, 23 documents). Preservation of the baseline firing level tracks $\cos(\hat{c}, \hat{m}_{10})$ monotonically.

layer	$w \cdot b_L$ (median)	b_n (median)	resting	ref-dominant	$\rho(\text{resting, duty})$	$\rho(\text{random dir})$
6	−1.535	−0.027	−1.568	99.98%	0.943	0.058
10	−1.226	−0.022	−1.250	99.99%	0.959	0.072
14	−1.091	−0.021	−1.115	99.96%	0.960	0.085
22	−1.130	−0.018	−1.149	99.94%	0.954	0.073

Table 3: Exact resting decomposition of gate pre-activations (Phi-2, 19 documents). “Ref-dominant” is the fraction of gates whose resting inhibition is carried mostly by $w \cdot b_L$ ($|w \cdot b_L| > |b_n|$). The last column repeats the rank correlation with a random direction substituted for b_L at matched norm (median of 50 draws, $p_{95} \leq 0.192$, maximum 0.223; from a 39-document run, which reproduces the $\rho(\text{resting, duty})$ column at 0.946–0.964). The $\rho(\text{resting, duty})$ column is near-tautological read alone; it is the *contrast* between the last two columns that carries direction specificity. 95% document-bootstrap CIs (2000 resamples): ref-dominant stays $\geq 99.89\%$ at every layer (widest [99.89, 99.98] at L22), and ρ is [0.936, 0.945]/[0.951, 0.959]/[0.953, 0.959]/[0.945, 0.952] (L6/L10/L14/L22; percentile intervals, and resampling biases ρ slightly low, so a point estimate can sit just above its interval). The 48–56× quoted in the text is the ratio of median magnitudes, $|w \cdot b_L|$ against $|b_n|$ (the latter 0.027/0.023/0.023/0.021), which differ slightly from the signed medians shown. The *resting* column is a per-gate median and does not equal the sum of the $w \cdot b_L$ and b_n medians, since a median does not distribute over addition across a heterogeneous gate population.

gain-boost alternative is excluded the same way: a random-direction replacement leaves the pre-norm magnitude in place, so the LayerNorm gain is the same as clean, yet firing releases as if the projection had been removed, while layer 18’s direction, under the same norm, preserves baseline firing. One further observation: the shuffled replacement’s KL (0.95) *exceeds* removal’s (0.52); transferring the projection onto a wrong direction injects a coherent error signal on top of losing the reference.

5 Implementation: the threshold is the coupling

For a gate n with read row w (a row of Phi-2’s $\mathbb{F} \subset \mathbb{1}$) and bias parameter b_n , the *resting* pre-activation decomposes exactly: $\langle w \cdot x_{\text{in}} + b_n \rangle = w \cdot b_L + b_n$, since $\langle x_{\text{in}} \rangle = b_L$ by definition. If gates threshold against the carried reference, training should have moved the operating-point inhibition into the coupling $w \cdot b_L$ rather than the explicit bias parameter b_n .

Table 3: the reference coupling carries the resting inhibition at 48–56× the explicit bias parameter (ratio of median magnitudes) for more than 99.9% of gates at every tabulated layer (the shallowest layer measured, L2, is at 98.5%), and the resting pre-activation rank-predicts each gate’s duty cycle at $\rho \approx 0.95$.

The $\rho \approx 0.95$ is near-tautological on its own, and the next paragraph derives its null; what survives is the narrower claim that the population’s operating point is set along this direction and not by the bias parameter. In the twin, $w \cdot b_L$ has no shared sign (median ≤ 0.01 , half the gates on each side, typical magnitude 0.5) and duty is ≈ 0.5 ; its bias parameters are exactly zero at init, so the informative twin statistic is that the population has no shared resting inhibition. This is the classical bias-as-weight-to-a-constant-input construction, discovered in the weights: the model implements thresholds as couplings to a constant it itself carries. It also explains the dose-response of §3 without further assumptions: scaling the stream’s projection on $\hat{m}_L$ scales the operating-point term of every gate simultaneously.

What the duty correlation does and does not show. Since $b_L := \langle x_{\text{in}} \rangle$, the resting term is the mean pre-activation, and duty is the fraction of that same distribution above zero. A rank correlation between the mean of a distribution and $P(> 0)$ is high for any family with comparable spreads, so $\rho \approx 0.95$ may be a property of the construction rather than of b_L , and the twin cannot adjudicate: with resting ≈ 0 for every gate it has no spread to correlate. We therefore built the null the twin does not provide. Permuting the resting values across gates (each gate keeps its own centred pre-activation distribution but receives an offset carrying no information about b_L) still yields $\rho = 0.87\text{--}0.93$ (L6/L10/L14/L22), which is 92–97% of the measured value. Injecting offsets drawn from the trained resting distribution into the *twin*, a model with no coupling at all, yields $\rho = 0.984$, higher than the trained model, because the statistic is governed by how homogeneous the per-gate spreads are (twin coefficient of variation 0.056 against the trained model’s 0.15–0.20) rather than by the reference. Consistently, with each gate’s resting term divided by its own pre-activation standard deviation σ_n , $\rho(\text{resting}/\sigma_n, \text{duty}) = 0.987\text{--}0.994$ exceeds the raw figure. The duty correlation is a location fact and we do not rest anything on it.

Two statistics do survive that test. First, direction specificity: a random direction at the same norm as b_L , substituted into the same decomposition, orders duty at $\rho = 0.058\text{--}0.085$ (median over 50 draws; $p_{95} \leq 0.192$, maximum 0.223) against the reference’s 0.946–0.964. The operating point is set along this specific direction, the static counterpart of the replacement panel of §4. Second, the coupling is largely norm-scaled: the read-weight norm alone orders duty at $\rho = -0.878$ to -0.896 , as the per-gate anatomy below implies (the coupling is a narrow, nearly fixed fraction of each row’s norm), and in the twin the same statistic is null (-0.010 to $+0.007$), so the norm-to-duty relation is itself trained. The read-weight norm accounts for most of the duty ordering; what b_L contributes is the part left after that norm-only ordering.

Per-gate anatomy. Factoring each fc1 row against the unit reference, $w = c_n \hat{b}_L + w_{\perp}$, the coupling is a small slice of read norm: $|c_n|/\|w\|$ has median 0.07–0.12 across layers against a random-twin floor of 0.013, thinning toward the readout, so ≈ 0.99 of the row remains free for content. The coupling alone rank-predicts duty as well as the full resting inhibition ($\rho(c_n, \text{duty}) = 0.943\text{--}0.959$, matching Table 3), so b_n adds nothing to the ordering; both figures inherit the caveat above. On its own b_n also orders duty ($\rho = 0.81\text{--}0.83$), but it adds nothing once the coupling is known. Per gate, the coupling takes 0.5–1.4% of the read weight’s energy and fixes where the gate sits while leaving its content direction almost untouched; $c_n = w \cdot \hat{b}_L$ is both the gate’s (negative) read of the reference and the term that sets its resting operating point.

The coupling over training. The decomposition above compares two endpoints. Pythia-410M-deduped, whose GELU gates carry an explicit bias and whose checkpoints are public, lets the two static terms be tracked step by step (four mid-stack layers). b_n is exactly zero at initialisation (Pythia zero-inits biases) and stays ≤ 0.04 in magnitude the entire run, while $|w \cdot b_L|$, after an initial drop ($\sim 0.25 \rightarrow 0.12$ by

model	gate (bias?)	resting med.	ref-dom.	$\rho(\text{rest, duty})$
GPT-2-medium	c_fc (yes)	−1.52 to −0.99	99.4–99.7%	0.91–0.97
Pythia-1.4B	h_to_4h (yes)	−1.08 to −0.74	> 99.9%	0.96–0.98
Qwen2.5-1.5B	gate_proj (no)	−4.14 to −0.37	—	0.62–0.99
TinyLlama-1.1B	gate_proj (no)	−0.21 to −0.07	—	0.97–0.99

Table 4: Cross-model resting decomposition (30 documents, four mid-stack layers per model at depth fractions 0.2–0.65; ranges span those layers). “Ref-dom.” is the fraction of gates whose resting inhibition is carried mostly by $w \cdot b_L$ rather than the bias parameter (vacuous for the bias-free SwiGLU gates; for these the informative statistic is the negative resting coupling and its duty-prediction). Where a bias exists, $|w \cdot b_L|/|b_n|$ is 15–18 $\times$ (GPT-2) and 38–43 $\times$ (Pythia). Every model’s random-init twin is flooded: $|\text{resting median}| \leq 0.01$, duty ≈ 0.5 .

step 128), builds through the step 512 $\rightarrow$ 1000 window (0.16 $\rightarrow$ 0.38), reaches 0.84–0.95 by step 4000, peaks at step 8000 (16000 at the shallowest layer), and relaxes partially thereafter. The reference coupling carries the resting inhibition of $\geq 99.6\%$ of gates at every checkpoint after initialisation, where b_n is exactly zero: there is no phase in which the explicit bias leads and is overtaken. The network does not route the operating point through the bias parameter at any stage; it builds the coupling directly. Duty falls 0.51 $\rightarrow$ 0.16 over the same build and $\rho(\text{resting, duty})$ stays 0.91–0.998 throughout. The time-course confirms the endpoint comparison, and it places the onset in the same window as §7 through a different statistic.

Cross-model replication. The resting decomposition is static (weights and mean activations only), so it can be re-run cheaply on other families against their twins. Table 4 does so on four more architectures, spanning both gate types: GELU MLPs with an explicit gate bias (GPT-2-medium, Pythia-1.4B) and bias-free SwiGLU gates (Qwen2.5-1.5B, TinyLlama-1.1B). In every family the resting pre-activation is negative (gates held off by the carried reference) and rank-predicts duty at $\rho \geq 0.91$ at every sampled layer but one (below), with the twin flooded (resting ≈ 0 , duty ≈ 0.5) throughout. Every ρ in this paragraph and in Table 4 inherits the caveat above and is reported as description; what carries the cross-family claim is the sign and magnitude of the resting term relative to b_n , and the random-direction control was run on Phi-2 only. Where a gate bias exists, the reference coupling dominates it for $> 99\%$ of gates at 15–43 $\times$ the bias parameter; where the gate is bias-free, the coupling alone carries the entire negative resting term. The one sub-threshold ρ is Qwen’s shallowest sampled layer ($L5$, $\rho = 0.62$, a strongly suppressed near-off regime, duty 0.004); every deeper layer is $\rho \geq 0.96$. Magnitude varies (TinyLlama’s resting inhibition is an order of magnitude weaker than GPT-2’s).

A counteracting boundary case. Extended across the eight-model GELU/ReLU/SiLU scan, the static decomposition finds the mechanism in all six non-OPT models (five families; ref-dominant 99.4–100% where a bias exists, $\rho \geq 0.91$ except at Qwen’s shallowest layer) and not in the two OPT models (opt-350m, opt-1.3b): their resting inhibition is 40–130 $\times$ weaker, bias-dominated, and rank-predicts nothing ($\rho = 0.000$, as follows from a resting term near zero). A norm decomposition explains why. Writing $w \cdot b_L = w \cdot (b_L - \beta) + w \cdot \beta$ (residual-carried mean plus LayerNorm bias), the residual-carried part is primary wherever the mechanism is present: it dominates 91–99% of gates in the GELU LayerNorm models and equals the resting term by construction in the bias-free RMSNorm models, with β a 7–26% augmentation. So against the LayerNorm bias, the §2 attribution of the reference to a residual-carried mean holds; the module biases are separated next. OPT shows what the mechanism requires: it builds a large

residual-carried inhibition comparable to GPT-2’s but trains an almost exactly opposing LayerNorm-bias excitation ($|w \cdot (b_L - \beta)| \approx |w \cdot \beta| \approx 0.5\text{--}0.6$, opposite sign), netting $w \cdot b_L \approx 0.006$: the reference is present but cancelled at the operating point. ReLU-versus-family stays formally confounded (every ReLU model sampled is an OPT).

How much of the reference is a parameter. The decomposition above separates the residual-carried mean from the LayerNorm bias, but it counts the accumulated *module* biases (each block’s attention-output and MLP-output bias) inside the residual-carried part, though they are input-independent, added unconditionally whatever the input. Separating them changes the accounting materially. Projecting β and every upstream module bias onto the reference (the module biases enter as a constant vector divided by the per-position σ , so the constant is computable from weights alone), the input-independent share of the reference component tracks architecture exactly across the replication set: 41–47% (Phi-2), 10–32% (GPT-2-medium), 52–84% (Pythia-410M), 26–40% (Pythia-1.4B), and *exactly* 0% in all three bias-free RMSNorm models (Qwen2.5-1.5B, TinyLlama, SmolLM2), which have neither β nor output biases and must therefore assemble the same reference entirely from computed writes. OPT is again the informative case: its share is *negative* (−16 to −8% in opt-350m), the same cancellation seen in the paragraph above from the parameter side. So the mechanism holds across the non-OPT architectures while its *ingredients* vary: how much of the reference is carried in the stream versus in the parameters is a proportion the architecture sets, and the span is wide, so no single figure should be quoted for it.

6 How the reference persists, and the cost of losing it

Attention carries the steady feed. Decomposing each block’s output into its component along $\hat{m}_L$ and the orthogonal remainder, and measuring the constant share of each (the product of its means over two halves of the corpus, as a fraction of its mean square): at L10, the attention block’s $\hat{m}_L$ channel has constant share 0.452 against 0.014 for its orthogonal channel, a $33\times$ dissociation; the block outputs a steady reference feed while its content channel fluctuates. The MLP shows no such dissociation (0.003 vs. 0.003): unlike attention it carries no steady reference feed, though its time-averaged write is reference-aligned (§2). In the twin both channels are largely constant (0.96 vs. 0.86; an untrained block is a mostly-constant filter), so the dissociation is a trained property. Together with §2: per-layer mean writes build the reference; attention sustains it in operation.

Restoration. A single-site removal is local in depth: downstream firing is undisturbed. After removal at L6 (§3), $\cos(\text{resid}, \hat{m}_L)$ falls by 0.16 at L10 and L14 and by 0.10 at L18, and recovers to baseline by L26–L30: the stream re-accumulates its mean over $\sim 12\text{--}16$ layers, consistent with the sources of §2 (every layer’s mean write is reference-aligned, so the reference restores itself by accumulation). Deeper removal sites restore more slowly.

Co-removal. Single-site removal understates the dependence because the stream heals. Removing the projection at *all* measured layers ≥ 6 simultaneously releases firing at every depth (most at the shallowest site, still +0.075 at L18) and costs $\text{KL} = 4.38$, $\Delta\text{NLL} = +4.75$ nats, about $8.4\times$ the L10 single-site removal (Table 1, KL 0.52), while the twin under the identical intervention loses 0.65 KL with flat firing (document-bootstrap 95% CIs over 23 documents: KL [3.74, 6.05], ΔNLL [4.62, 5.03], disjoint from the twin). Late-only co-removal (≥ 14) is far milder (KL 0.55, CI [0.24, 0.90]): the dependence is concentrated mid-stack. The reference is a standing condition across the middle of the network, and no single layer carries the dependence alone.

7 Development

The operating point of §5 is built early in training, and by a token-local pressure. Three developmental settings, at increasing resolution, show this.

Pythia-410M: the reference forms in the induction window. On Pythia-410M-deduped public checkpoints (a different family and scale from the causal experiments; one seed), the reference coupling forms in the same early window as gate sparsification and the induction transition. In the step 512→1000 interval the induction-head score rises $0.03 \rightarrow 0.88$ (Olsson et al., 2022), the median duty cycle falls $0.42 \rightarrow 0.28$ (continuing to 0.16 by step 4000), and the reference share of drive deepens $-0.06 \rightarrow -0.34$ (to ≈ -0.75 by step 3000–4000); the static coupling $|w \cdot b_L|$ builds in the same window (§5). Pythia has no checkpoint between 512 and 1000, so this grid places the onset in that one interval but cannot order the three within it. Below it nothing is yet moving: across log-spaced steps 1–256 the induction score is at its floor, duty ≈ 0.50 , and the reference share ≈ 0 at mid-stack. The only earlier event is training’s first act on the MLP write, which destroys the constant the MLP writes at initialisation (the write’s constant share, as in §6, falls $0.40 \rightarrow 0.15$ by step 128). Massive activations follow a different early course in the same model family: absent at initialisation, they rise to an early peak in shallow and deep layers (Gallego-Feliciano et al., 2025). The trained reference is therefore built during training, and it is in place by the time the induction transition completes.

A 160M model: the operating point forms first. A finer developmental grid resolves the order the Pythia checkpoints cannot. In a 160M model (Pythia-160M architecture, a standard GELU MLP, trained here on the Pile with dense early checkpoints), the gate operating point forms well before the induction transition: median mid-stack duty falls from 0.50 to 0.075 by step ≈ 800 (and lower thereafter), and the resting depth z (the resting pre-activation divided by its per-unit standard deviation σ_n , which puts the operating point in units of the gate’s own spread) reaches ≈ 1.5 over the same window, while the induction-head score is still near its floor and completes its rise only over steps ≈ 3000 –12000. Throughout, the operating point is the reference coupling (the mean-input term carries ≈ 0.99 of the resting pre-activation, as in §5), so it is the mechanism of §5 that is in place first. (Absolute step counts are not comparable to Pythia-410M’s, which uses a different schedule and $\sim 100\times$ more tokens per step; the developmental *order* is the point.)

An attention-free model: the operating point is token-local. The operating point forms without any cross-token computation. Training the same 160M architecture with attention disabled (every layer’s attention output projection zeroed and frozen, leaving a purely position-wise model that predicts each token from the current token alone, a bigram map) reproduces the same development. In a matched pair of runs differing only in whether attention is enabled (identical script and schedule, to step 3000), the attention-free model’s mid-stack duty settles at 0.077 against 0.067 for the attention-carrying one (a $\approx 15\%$ higher floor) and its operating point is slightly shallower ($z \approx 1.3$ against ≈ 1.5), while the reference coupling is unchanged (the mean-input term carries ≈ 0.99 of the resting pre-activation in both). Attention makes sparsification marginally faster and deeper, but the operating point forms under the token-local next-token prediction task alone, whether or not the model can read other positions (one seed per arm).

Reading. Across the three settings (each one seed) the operating point is an early, token-local component of a trained MLP: the network sets its gates against the carried reference as it learns the base next-token

map, in the same early window in which cross-token circuits mature at scale. Activation sparsity is a known endpoint property (Li et al., 2022; Szatkowski et al., 2025), and Shvetsov et al. (2026) give an optimisation account of early training that drives weights negative (shown in GPT-nano, for weights that read positive activations); the reference coupling is how that endpoint is reached here, and the pressure that reaches it is present even in a model with no context to read.

8 Discussion

Representational analyses commonly subtract the corpus mean, and for many questions that is the right step: mean-centering for analysis leaves the model’s computation untouched. The results here say what the subtracted component is. It is the reference against which the gate population’s default-off state is defined; removing it from the forward pass costs KL 0.27–1.43 at a single site in Phi-2 and 4.38 when it is removed at every measured layer from L6 on, and the damage falls on the least certain predictions (one site, 16 sentences). An account of gate behaviour built on mean-centred activations has set aside the term that decides where the gates rest, and needs it back when the question is why a unit is off. The same mechanism appears in sparse autoencoders: a feature’s encoder pre-activation is shifted by its alignment with the activation mean, so features anti-aligned with it start with negative pre-activations and can stay dead, and centering the autoencoder’s input removes the effect (Simon et al., 2026); there the effect is large in vision and protein models and small in the language models tested. On the architecture side, PaLM dropped biases from its dense kernels and layer norms for training stability (Chowdhery et al., 2022); the decomposition suggests why this need not cost anything on the gate side, since outside OPT the operating point never lived in b_n : the network builds the coupling from early training even when a bias is available (§5, the coupling over training), and the bias-free RMSNorm families assemble the whole reference from computed writes (§5).

The coupling can also be followed into MLPs built from gated linear units (GLU), where the read stage has a gate branch and a value branch. There it sits in the gate branch, and a gate unit’s coupling to b_L places it on a continuum from a thresholded switch to a bilinear multiplier of its two reads; that analysis is reported separately (Tuomi, 2026).

9 Related work

Massive activations, sinks, outlier dimensions. Attention sinks are initial tokens that receive strong attention regardless of their content (Xiao et al., 2023), and massive activations are near-constant residual components that Sun et al. (2024) describe as “indispensable bias terms”, acting as an implicit bias in self-attention; Owen et al. (2025) find that this does not hold uniformly across architectures, since suppressing massive activations is not harmful in every model. Chen et al. (2026) show that spike-carrying tokens become constant vectors after normalisation that act as structural biases inside attention, and later work ties sinks to compression valleys (Queipo-de Llano et al., 2025), to outlier-driven rescaling essential for training (Qiu et al., 2026), and to the damping of transient amplification across depth (Hengyu, 2026). Most recently, Sun et al. (2026) give a functional split: massive activations as near-constant hidden components acting as implicit parameters, and attention sinks as local attention modulators; they identify the pre-norm configuration as what couples the two, decoupling them under ablation. Outlier dimensions track token frequency (Puccetti et al., 2022), go with high-magnitude LayerNorm scaling parameters (Kovaleva et al., 2021), and can dominate cosine similarity (Timkey and van Schijndel, 2021), and Macocco et al. (2025) show that last-layer outlier dimensions act as a constant positive logit contribution toward frequent tokens. The object studied here is in the same family but the measured function differs in

site and form: a *direction* (not a coordinate set), read at the MLP gate inputs mid-stack (not at the output logits), that carries the resting inhibition of the gate population by exact decomposition, is direction-specific under replacement, and is assembled by distributed aligned writes. The context-resolved removal of §3 does place the two objects in overlapping *functional* territory, since removing the reference costs most at function-word and high-entropy positions; we take the separation to be one of object and site rather than of function, and the geometric comparisons below are what carry it.

We ran the geometric comparison against these objects on Phi-2. Mid-stack, $\cos(b_L, \cdot)$ against the first-token direction at its emergence layer (located here at layer 21, peak coordinate magnitude ~ 1250) is 0.04–0.17, and against the residual-sink and massive-coordinate directions ≤ 0.27 : the objects are distinct. Shi et al. (2026) note that such a first-token massive activation “can serve as a stable and shared global reference vector” for attention, but argue that its rigidity limits input-dependent attention. The separation is trained: in the random-init twin the reference is the attention-sink direction ($\cos = 0.99$, no massive activation), and training rotates the reference off the sink. On outlier dimensions: b_L carries 10–23% of its squared norm on the twenty largest last-layer outlier coordinates against an isotropic share of 0.8%, a real but minority overlap that grows toward the readout; its own dominant coordinates are largely disjoint from the outlier set (top-20 Jaccard 0.05–0.14). Mid-stack, b_L is a distributed direction with a minority frequency-correlated component, distinct from the axis-aligned outlier set, and it should not be conflated with the sink objects of Qiu et al. (2026) or Shi et al. (2026). This distinctness is a property of Phi-2. In models with strong massive activations the mean concentrates on the massive coordinates: in Qwen3-architecture models trained from scratch, Cao et al. (2026) find that the dominant activation outliers late in training are largely induced by a rank-one mean bias, and in the SwiGLU models of Tuomi (2026) the reference’s largest coordinates coincide with the massive-activation coordinates.

Distinctness under a variance-aware metric. The comparisons above use Euclidean cosines, and Ying et al. (2026) note that the Euclidean cosine treats all dimensions equally, although only the dimensions along which the data vary matter, and a few such dimensions can dominate it (Timkey and van Schijndel, 2021). That objection is pointed here, because the objects we compare against (the residual sink and the massive activations) are those high-variance coordinates, so an equal-weight cosine could be understating an alignment with them. We therefore recomputed the comparisons under variance-aware metrics, on the same directions and corpus, so that any change is attributable to the metric. In the space b_L is defined in and gates read (the covariance of the layer-normalised MLP input), the covariance-weighted cosine of Ying et al. (2026) places b_L against the sink, first-token and massive directions at -0.23 to $+0.01$, against an empirical random-pair null of 0.03 median and 0.11 at the 95th percentile; restricting to the top-512 principal subspace gives -0.41 to -0.14 . The positive control, the covariance-weighted $\cos(b_L, m_L)$, is 0.91–0.95 throughout. Under the metric the objection prescribes, the objects show no alignment, and some are mildly opposed. The instrument is validated on the twin, where b_L is the attention-sink direction: there every metric returns 0.83–0.99. Both metrics show one trend: the overlap with the sink grows toward the readout ($0.12 \rightarrow 0.27$ Euclidean over L6–L18, $0.08 \rightarrow 0.33$ in the principal subspace of the raw-residual covariance), in step with the outlier-coordinate mass reported above.

Bias as coupling. Treating bias as a weight onto a constant input is the classical homogeneous-coordinates construction. A weight-based taxonomy of gated neurons (Gerstner and Schütze, 2025) and an activation-based reading of MLP units as binary routers (Balogh, 2026) characterise what individual gates do, and massive activations have been read as carried constants with a bias-like role (Sun et al., 2024, 2026). The observation here joins these: a trained transformer implements the homogeneous-coordinates construction spontaneously, with the constant supplied by the network’s own carried mean

rather than by a dedicated input. Sun et al. (2024) also report the trained converse on the attention side: augmenting self-attention with explicit learnable bias parameters yields GPT-2 models that do not form massive activations (the same remedy is ineffective in some other architectures; Owen et al., 2025): given the parameter, the network uses it instead of building a carried constant. OPT plays the same role on the gate side here (§5): the parameter-based strategy exists, and most families do not take it.

10 Limitations

- **Replication scope.** The static decomposition (§5) is shown on the eight-model scan and is absent in OPT, where an opposing LayerNorm bias cancels the carried reference; the ReLU-versus-OPT-family confound (every sampled ReLU model is an OPT) is not separated by matched-nonlinearity twins. The causal dose-response and the entropy release side replicate on two further families (Pythia-1.4B, Qwen2.5-1.5B); the replacement panel (§4, one site, L10) and co-removal (§6) are Phi-2-only. The development results (§7) are trend-level and single-seed: the Pythia-410M checkpoint time-course is one seed (15–30 documents), and the finer-grid and attention-disabled results are each one 160M model trained here, one seed per arm. Two further results are single-model and modest- n : the depth profile of the mean’s share of the stream (§2) and the context-resolved removal (§3, one site, 16 sentences).
- **Cross-layer geometry is family-dependent.** The rotating-band description (§2), and the transfer measurement that goes with it, are Phi-2 only; the cross-layer cosine is heterogeneous across the replication families: $\cos(\hat{m}_L, \hat{m}_{L'})$ between mid-stack layers is high in GPT-2-medium (0.95) and Qwen2.5-1.5B (0.96), modest in TinyLlama-1.1B (0.74), and below its own twin floor in Pythia-1.4B (0.41 vs. twin 0.67). The static coupling replicates in every non-OPT family and the causal dose-response in all three models tested; the single-carried-direction geometry does not replicate, and the paper does not claim it.
- **Which statistics carry which claim.** The rank correlation between the resting term and duty is near-tautological on its own (§5), because the resting term is the mean pre-activation and duty is $P(> 0)$ of the same distribution; we report its nulls and rest nothing on it. What the static decomposition establishes is the sign and magnitude of the resting term against b_n , and, on Phi-2, direction specificity against a matched-norm random control. The functional claims rest on the causal tests (§3, §4, §6), which do not use the correlation at all. The duty ordering is also largely norm-scaled ($\rho = -0.89$ from the read norm alone), so the reference’s contribution to the ranking is the residue above that, which we have not decomposed further.
- **Within-layer atomicity.** b_L is a corpus mean, rank-1 by construction; the token-class structure it hides (§2) carries only 2–4% of the variance around it, so the rank-1 treatment is a good approximation for the operating-point claims, and every causal result here concerns only the rank-1 mean.
- **Operating-point asymmetry.** The entropy response to weakening the reference (§3) rests mainly on the release direction, since the amplification side is shallow in Phi-2 and varies across families.
- **Manipulation scope.** The reference is the corpus mean; a position-resolved reference is a refinement not tested. Manipulations leave positions < 16 untouched, so attention re-import from the sink region is possible. α -scaling changes the LayerNorm input distribution, so all downstream effects pass through LayerNorm renormalisation; the norm-matched control shares this property, and the

norm-preserving replacement panel (§4) excludes a pure gain-boost account at L10, the one site where it was run. The function-word replacement is one naive construction of a content-direction control.

- **Geometric comparisons are activation-proxy.** The comparisons against first-token massive activations, the residual sink, and last-layer outlier dimensions (Related work) use activation-based proxies of each object on one model; a weights-only unification against each paper’s exact construction is not attempted. Two caveats attach to the variance-aware recomputation reported alongside them. Weighting instead by the *raw residual* covariance is uninformative here, because the comparison objects are themselves the high-variance coordinates: that metric returns nearly the same value for all three proxies (within 0.004), and its own random-pair null reaches 0.52–0.73 at the 95th percentile, so it cannot discriminate. And an inverse-covariance (whitening) variant fails its own positive control at this sample size (8,868 positions for a 2560-dimensional covariance), ranking $\cos(b_L, \text{sink})$ above $\cos(b_L, m_L)$ at the weakest regularisation (at L6 and L18) and holding the positive control to only 0.70–0.80 at the others; we report it as excluded. The recomputation is Phi-2-only, like the comparisons it checks.

11 Reproducibility

Scripts and JSON artifacts (public checkpoints, fixed seeds):

census_background_reference.py (§2, §3), census_background_identity.py (§4, co-removal), census_reference_mechanism.py (§5, §6), census_checkpoint_trajectory.py, census_formation_microscopy.py and census_formation_prelude.py (§7, Pythia-410M; the last is the sub-512 resolution check), induction_gate_coreg.py (§7, the 160M finer-grid duty-vs-induction time-course) and train_noattn_sparsification.py (§7, the 160M model and its attention-disabled variant), census_reference_geometry.py (geometric comparisons, Related work), census_reference_geometry_mahalanobis.py (variance-aware recomputation of those comparisons, with per-metric empirical nulls), census_reference_operating_point.py (§2, the depth profile of the mean’s share of the stream and the post-norm reference), census_reference_statistic_nulls.py (§2, §5: the permuted-offset and twin-injection nulls for the duty correlation, and the cross-layer cosine and transfer matrices), census_reference_duty_controls.py (§5, the random-direction, read-norm and bias-only controls), atlas_backbone.py (§3, context-resolved removal), census_noise_floor.py and census_noise_zero_mean.py (§3, the matched-energy noise floor and the direction-matched injection), census_crossmodel_causal.py (cross-model dose-response and entropy response), census_bias_migration_scale.py (Table 4), census_bias_migration_formation.py (coupling time-course over checkpoints), census_bias_migration_nonlinearity.py (eight-model scan, OPT boundary), census_reference_norm_decomp.py (residual-versus-LayerNorm-bias decomposition), census_reference_position_shape.py (§2, the prose, code and repeated-token reference cosines), census_pass1_scan.py and census_drive_ledger.py (the duty and reference-share figures of the introduction), census_readweight_factorization.py (per-gate anatomy), census_reference_variance_sources.py (what holds the fraction, §2), census_reference_mixture.py (token-class structure, §2), census_reference_bias_and_depth.py and census_reference_bias_share_xmodel.py (the parameter share, §5),

`census_reference_text_statistics.py`, `census_sigma_dose_response.py` and `census_duty_at_task_position.py` (the input-side demonstration and its scope, §3), `census_reference_horizon.py` (the sequence-position horizon). Code and data are available at <https://github.com/EvidentSolutions/llm-interp/tree/main/background> and archived at <https://doi.org/10.5281/zenodo.21498411>.

Use of AI Assistants

Large language models were used as assistive tools for coding, running experiments, and drafting text. All research questions, experimental design, and reported claims were directed and verified by the author, who takes full responsibility for the content.

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